TICKET ID: WEEK5-LAB-ESC-001

ESCALATED TO TIER 2

L1 TO L2 SOC ESCALATION TICKET

DG Interns Hub — Security Operations Center | Week 5 Internship Assignment

1. TICKET HEADER & INCIDENT INFORMATION

Ticket ID:	WEEK5-LAB-ESC-001	**Alert Name:**	Linux Authentication Failure Investigation
Affected Host:	ubuntu-soc	**Affected User:**	gunjan
Log Source:	/var/log/auth.log	**Splunk Index:**	main
Sourcetype:	linux_secure	**MITRE ATT&CK:**	T1110 — Brute Force (Potential)
L1 Analyst:	SOC Analyst Intern	**Handover Date:**	2026-11-06 (Lab Execution)

2. SEVERITY & JUSTIFICATION

LOW SEVERITY — LIKELY BENIGN MISTYPE

Justification: A total of 3 authentication failure records were identified for local user gunjan on host ubuntu-soc. All failures occurred via local PAM service handlers (gdm-password for desktop unlock and sudo for command execution). Zero remote connection attempts (e.g., SSH / sshd) or external attacker IP addresses were observed. Furthermore, a successful local session-open event was logged approximately 14 seconds following the failure, strongly indicating a routine user password mistype rather than malicious adversary activity.

3. ALERT DESCRIPTION

This alert was triggered by automated SPL detection rules identifying clustered authentication failure logs within Linux /var/log/auth.log ingested into Splunk Enterprise under index=main sourcetype=linux_secure. The search flags localized authentication failure strings ("authentication failure" or "Failed password") to monitor for potential password spraying or credential guessing attempts targeting Linux system accounts.

4. INDICATORS & EMPIRICAL LAB EVIDENCE

INDICATOR CATEGORY	OBSERVED LAB DATA / VALUE	CONTEXT / TRIAGE EVALUATION
Splunk Event Count	4 Raw Events (2-3 Aggregated Rows)	Low volume failure cluster within evaluated time window.
Failure Breakdown	2x gdm-password, 1x sudo	Local desktop GUI unlock & local privilege escalation attempts.
Target Account	gunjan	Standard local user account on host ubuntu-soc.
Remote Source IP	**Not Observed**	All events generated internally; no remote IP recorded in logs.
Destination IP	**Not Available** (Local loopback/host)	Host-level local PAM authentication.
Session Correlation	Session opened ~14s post-failure	Indicates immediate successful user authentication following mistype.

5. INVESTIGATION TIMELINE

- **2026-11-05 (Lab Timeframe):** Initial authentication failures generated on `ubuntu-soc` for user `gunjan` across GNOME display manager (`gdm-password`) and `sudo`.
- **2026-11-05 (+14 Seconds):** Local session-open event recorded for user `gunjan`, confirming successful authentication.
- **2026-11-06 12:38 IST:** L1 Analyst executed Splunk Search 01 (Brute Force Detection), returning 4 raw events / 2 aggregated statistics rows.
- **2026-11-06 12:52 IST:** L1 Analyst executed Splunk Search 06 (Auth Failure Analysis), confirming 3 failures across `gdm-password` and `sudo` processes.
- **2026-11-06 12:55 IST:** L1 Analyst conducted SSH-focused query check, returning 0 events. Ticket escalated to L2 for policy verification.

6. SPLUNK SPL QUERIES PERFORMED

Search 01 — Brute Force Detection Pipeline

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") | stats
count as failed_attempts by host, process, logname | sort - failed_attempts
```

Search 06 — Consolidated Process Analysis

```
index=main sourcetype=linux_secure ("authentication failure" OR "Failed password") | stats
count as failure_count values(process) as processes by host, logname | sort - failure_count
```

SSH Remote Vector Verification Check

```
index=main sourcetype=linux_secure process=sshd ("Failed password" OR "invalid user") | table
_time host src_ip user _raw
```

7. THREAT INTELLIGENCE ENRICHMENT STATUS

Status: NOT OBSERVED / NOT APPLICABLE

Because all authentication failure events originated locally on host `ubuntu-soc` via system PAM modules (`gdm-password`, `sudo`), **no external IPv4/IPv6 source address was present in the log payload**. Threat intelligence lookup (e.g., VirusTotal, AbuseIPDB, AlienVault OTX) could not be performed. No external IOCs or threat actor scores are associated with this alert.

8. L1 ANALYST ASSESSMENT

Based on empirical Splunk log analysis, the activity is classified as **Likely Benign (User Password Mistype)**. MITRE ATT&CK **T1110 (Brute Force)** is assigned strictly as a potential detection mapping and **does not represent confirmed adversary behavior**. The low failure count (3 failures), local execution context, absence of remote SSH attempts, and rapid subsequent session establishment (~14 seconds post-failure) support the conclusion of benign operator error during local lab activity.

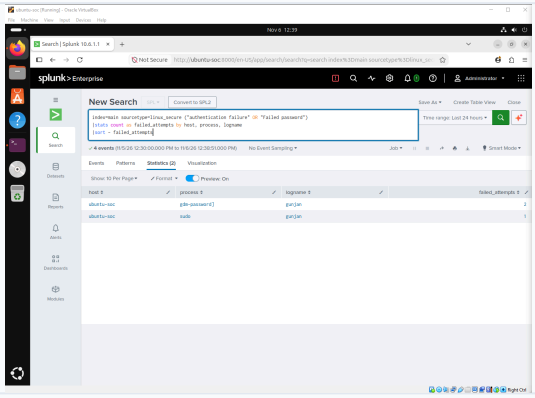
9. REASON FOR ESCALATION

Escalated to Tier 2 SOC in compliance with SOC Standard Operating Procedures (SOP) requiring formal L2 peer review for all authentication anomaly alerts. Escalation validates that no secondary stealth persistence mechanisms (e.g., unauthorized SSH key placement or backdoor account creation) were instituted surrounding the failure window.

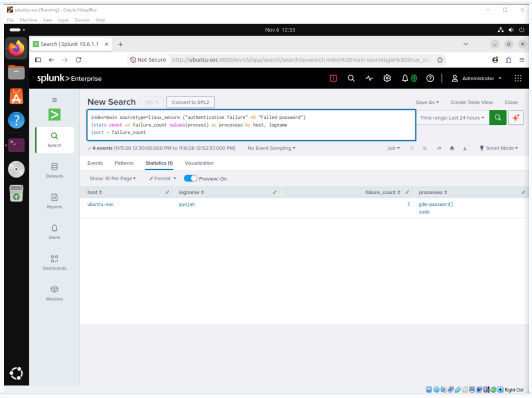
10. RECOMMENDED L2 INVESTIGATION STEPS

1. **Session Correlation Verification:** Inspect raw events immediately preceding and following the failure timestamps to verify the exact timeframe of the `pam_unix(system-auth:session): session opened` event.
2. **Persistence Audit:** Inspect `/home/gunjan/.ssh/authorized_keys` and `/etc/sudoers` to ensure no unauthorized persistence mechanisms were established.
3. **Log Integrity Check:** Confirm `/var/log/auth.log` continuity to ensure no log deletion or suppression occurred around the event timeframe.

11. EVIDENCE & SCREENSHOT REFERENCES



Ref 1: 01_Brute_Force_Detection.png (4 events / 2 rows)



Ref 2: 06_Linux_Auth_Failure_Analysis.png (3 failures)

12. FINAL HANDOVER NOTES

Handover Summary: Ticket `WEEK5-LAB-ESC-001` is formally handed over to Tier 2 SOC. All primary evidence from Splunk Enterprise 10.6.1 lab execution has been attached. Pending L2 review, L1 analyst recommends closing this ticket as **Benign / False Positive (User Mistype)** with no user account isolation required.